

SAMUEL FISHER

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EDUCATION AND EMPLOYMENT

ICMAT Madrid	2025 –
Severo Ochoa postdoctoral researcher in Bridson–Reid Lab	
University of Oxford	2020 – 2025
DPhil Mathematics	
Supervised by Prof. Dawid Kielak	
McGill University	2019 – 2020
MSc Mathematics	
Supervised by Prof. Daniel T. Wise	
McGill University	2016 – 2019
BSc Honours mathematics and physics	

PREPRINTS AND PUBLICATIONS

12. (with Pablo Sánchez-Peralta) *Coherent RFRS groups (with an appendix by Marco Linton)*, arxiv.org/abs/2603.16763
11. (with Pablo Sánchez-Peralta) *On the top-dimensional L^2 -Betti number of residually poly- $\mathbb{Z}$ groups*, arxiv.org/abs/2601.18594
10. *Novikov cohomology, finite domination, and cohomological dimension*, arxiv.org/abs/2510.14796
9. (with Dawid Kielak and Giovanni Italiano) *Virtual fibring of Poincaré-duality groups*, arxiv.org/abs/2506.14666
8. (with Sam Hughes) Appendix to *Profinite properties of Coxeter groups*, by S. Hughes, P. Möller, and O. Varghese, arxiv.org/abs/2505.08701
7. (with Kevin Klinge) *Dimension drop in residual chains*, Trans. Am. Math. Soc. **379**, no. 7 (2026)
6. *On the cohomological dimension of kernels of maps to $\mathbb{Z}$* , Geom. Topol. **30**, no. 1 (2026)
5. (with Ismael Morales) *The Hanna Neumann Conjecture for graphs of free groups with cyclic edge groups*, Compos. Math. **161**, no. 7 (2025)
4. (with Pablo Sánchez-Peralta) *Division rings for group algebras of virtually compact special groups and 3-manifold groups*, J. Comb. Algebra **10**, no. 1 (2026)
3. (with Sam Hughes and Ian J. Leary) *Homological growth of Artin kernels in positive characteristic*, Math. Ann. **389**, no. 1 (2023)
2. *Algebraic fibring of a hyperbolic 7-manifold*, Bull. Lond. Math. Soc. **55**, no. 3 (2023)
1. *Improved algebraic fibrings*, Compos. Math. **160**, no. 9 (2024)

INVITED TALKS

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| • TBD, Fibrations in group theory and topology, Oxford | September 2026 |
| • <i>Coherent RFRS groups</i> , Workshop on arithmetic (and non-arithmetic) groups, Maryland | July 2026 |
| • TBD, Groups and geometry in the southeast, Oxford | May 2026 |
| • Σ -invariants of coherent groups, Zaragoza Group Theory Seminar | May 2026 |
| • <i>Outer automorphism groups and the Zero Divisor Conjecture</i> , GTA Bilbao Algebra Seminar | April 2026 |
| • <i>Coherence and the vanishing of higher L^2-Betti numbers</i> | April 2026 |
| Manifolds and groups in Bologna IV | |
| • <i>3-manifold group algebras and division rings</i> | December 2025 |
| Workshop on infinite dimensional division algebras, SLMATH, California | |
| • <i>Novikov rings and cohomological dimension</i> , ICMAT Group Theory Seminar | February 2026 |

• <i>A Novikov ring short exact sequence</i> , Geometry and topology of polyhedral complexes (Mike Davis 75th birthday conference), Ohio State University	May 2025
• <i>Virtually free-by-cyclic RFRS groups</i> , McGill Geometric Group Theory Seminar	March 2025
• <i>Virtually free-by-cyclic RFRS groups</i> , Cambridge Geometric Group Theory Seminar	March 2025
• <i>3-manifolds in division rings</i> , Cambridge Junior Geometry Seminar	March 2025
• <i>Free-by-(poly)cyclic groups</i> , Bristol Algebra Seminar	November 2024
• <i>Maps to the circle, to the integers, and L^2-Betti numbers</i> University of Pisa Geometry Seminar	June 2024
• <i>Virtually free-by-cyclic RFRS groups</i> , Modern methods in infinite groups, Bristol	June 2024
• <i>L^2-Betti numbers and kernels of maps to $\mathbb{Z}$</i> Conference on homology growth and fibering, Bonn	May 2024
• <i>Virtual fibering of manifolds and groups</i> , Glasgow Geometry and Topology Seminar	March 2024
• <i>Kaplansky's zero divisor conjecture via embeddings into division rings</i> Southampton Pure Maths Colloquium	March 2024
• <i>Algebraic fibering and ℓ^2-Betti numbers</i> , McGill Geometric Group Theory Seminar	March 2023
• <i>Homology gradients of Bestvina–Brady groups</i> , Cambridge Junior Geometry Seminar	February 2023
• <i>Analogues of ℓ^2-Betti numbers in positive characteristic and homology growth</i> ICMAT Group Theory Seminar	January 2023
• <i>ℓ^2-Betti numbers and algebraic fibering</i> , ICMAT Group Theory Seminar	January 2023
• <i>Agrarian invariants and a positive characteristic version of Lück's approximation</i> Southampton Pure Maths Lunchtime Seminar	November 2022
• <i>Algebraic fibering and ℓ^2-Betti numbers</i> , KIT AG Topology Seminar	November 2022

AWARDS

NSERC Postgraduate Scholarship – Doctorate	2022 – 2025
ERC studentship	2020 – 2024
NSERC Canada graduate scholarship – Master's	2019 – 2020
Award for Outstanding Teaching Assistant	2019
NSERC Summer Undergraduate Research Award – Supervised by Prof. Jacques Hurtubise	Summer 2018
Undergraduate Summer Research Award – Supervised by Prof. Eyal Goren	Summer 2017
J.W. McConnell major renewable entrance scholarship	2016 – 2019
Royal Conservatory of Music Gold Medal	2010 – 2016
Top piano performance mark in Québec for grades 3, 4, 5, 7, 9, and 10	

TEACHING

University of Oxford

M2 Analysis I and II	Tutor for visiting students at Worcester	Hilary/Trinity 2024
M1 Groups and Group Actions	Tutor for visiting student at Worcester	Michaelmas 2024
Stipendiary Lectureship at St. Peter's College		2024 – 2025
Tutor for: M2 Analysis I, II, and III, and ASO Graph Theory		
B4.1 Functional Analysis I	Tutor for visiting student at St. Peter's	Trinity 2024
ASO Graph Theory	Tutor at St. Peter's	Trinity 2024
ASO Projective Geometry	Tutor at Queen's	Trinity 2024
M2 Analysis III	Tutor at Queen's	Trinity 2024
M2 Analysis II	Teaching assistant at Queen's	Hilary 2024
B3.1 Galois Theory	Tutor	Michaelmas 2023
B1.1 Logic	Tutor	Michaelmas 2023
Stipendiary Lectureship at Hertford College		Michaelmas 2022
Tutor for: A0 Linear Algebra and A2 Metric Spaces and Complex Analysis		
A3 Rings and Modules	Tutor for visiting student at St. Edmund Hall	Trinity 2022
A5 Topology	Tutor for visiting student at St. Edmund Hall	Hilary 2022

B1.2 Set Theory	Tutor	Hilary 2022
C2.2 Homological Algebra	Teaching assistant	Michaelmas 2021
A8 Probability	Tutor for visiting student at St. Edmund Hall	Michaelmas 2021
ASO Graph Theory	Tutor at Queen's and St. Anne's	Trinity 2021
C3.2 Geometric Group Theory	Teaching assistant	Hilary 2021
C3.1 Algebraic Topology	Teaching assistant	Michaelmas 2020
B7.1 Classical Mechanics	Teaching assistant	Michaelmas 2020

McGill University

Algebra I	Teaching assistant	fall 2019
Grader for: Hons. Analysis II (winter 2019), Hons. Analysis I (fall 2018), Hons. Applied Linear Algebra (winter 2018), Hons. ODEs (winter 2018), Advanced Calculus (fall 2017), Algebra I (fall 2017)		

Dawson College

Weekly lectures on problem solving techniques for mathematics competitions	winter 2021
Lecture on Cantor's diagonal argument for π -day	March 2021
Weekly zoom lectures on set theory	fall 2020
Weekly lectures on ODEs	winter 2020
Weekly lectures on multivariable calculus	fall 2019